

Test III

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Novel Prep

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Phase 3 · Mock Exam Training

Overview

1 Test III

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22 Module 2-style mock exam questions

Question 1

A packaging engineer is optimizing a cardboard sleeve whose width profile is modeled by the quadratic expression $ax^2 + 176x + c$, where a and c are positive constants. If $px + q$ is a factor of the expression,

$$ax^2 + 176x + c = (px + q)(rx + s),$$

where p and q are positive constants, what is the greatest possible value of ac ?

Answer 1

Final Answer: 7744

Question 2

A signal-processing filter uses a polynomial in the variable z^9 to tune resonance. In the given expression,

$$34z^{18} + bz^9 + 30,$$

b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

Answer 2

Final Answer: 1021

Question 3

An optics lab rewrites a focusing model into paired quadratic factors in x^2 . The expression

$$6x^4 + 17x^2 + 7$$

can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive non-integers.

What is the value of $a + c$?

Answer 3

Final Answer: $\boxed{\frac{17}{2}}$

Question 4

A materials scientist factors a stiffness polynomial before running a simulation. The given quadratic function

$$54x^4 + 219x^2 + 105$$

has factors in the form $(k)(ax^2 + b)(cx^2 + d)$. If a, b, c, d, k are integers, what is the smallest possible value of ab ?

Answer 4

Final Answer: 14

Question 5

A drone's altitude path over a short flight is modeled by a parabola in the xy -plane. The equation of a parabola is written in the form

$$y = ax^2 + bx + c,$$

where a, b, c are constants. When graphed in the xy -plane, the parabola has vertex $(-1, 4)$ and does not intersect the x -axis. Which of the following could be the value of $a - b - c$?

- A. 2
- B. 4
- C. -1
- D. -6

Answer 5

Correct Answer: D

Question 6

A fountain's water arc is modeled by a downward-opening quadratic with vertex $(1, 32)$. A quadratic function with vertex $(1, 32)$ can be written as

$$-ax^2 + bx + c \quad (a, b, c \text{ are all positive integers}).$$

What is the maximum value of b ?

Answer 6

Final Answer: 62

Question 7

A robotics team measures that a test cart's speed is increasing at a rate of 7.7 meters per second squared. What is this rate in **miles per minute squared**, rounded to the nearest tenth? (Use 1 mile = 1609 meters.)

- A. 0.3
- B. 17.2
- C. 206.5
- D. 209

Answer 7

Correct Answer: B

Question 8

A bacteria culture in a bioreactor is modeled by an exponential growth function. The function f is defined by

$$f(x) = ab^{x/n},$$

where a, b, n are constants, and b, n are integers. If $f(2) = 6$ and $f(4) = 150$, what is the value of $f(5)$?

Answer 8

Final Answer: 750

Question 9

A roller-coaster design uses a parabolic track segment $y = f(x)$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(4, -7)$, contains the point $(3, -9)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

Answer 9

Final Answer: 195

Question 10

An economist fits a quadratic revenue model $y = f(x)$ that breaks even at two production levels. The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a, b, c are constants. The graph of $y = f(x)$ passes through the points $(9, 0)$ and $(-2, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- A. 8
- B. 7
- C. -6
- D. -12

Answer 10

Correct Answer: D

Question 11

A bridge support cable clearance is checked using polynomial factors in x . Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 22x + 14b$
- C. $3x^2 + 30x + 14b$
- D. $3x^2 + 37x + 14b$

Answer 11

Correct Answer: D

Question 12

A survey crew measures a triangular lot XYZ for a small park. In triangle XYZ , the measure of angle Y is 38° . The length of XY is 12 meters, and the length of YZ is 5 meters. What is the area, in square meters, of triangle XYZ ?

- A. 30
- B. 60
- C. $30 \sin 38^\circ$
- D. $60 \sin 38^\circ$

Answer 12

Correct Answer: C

Question 13

A quality team selects two random samples to estimate the average lifespan of a street-lamp bulb (in hours). Based on the first sample, the estimated average lifespan is 800 hours, with a margin of error of 20 hours.

Based on the second sample, the estimated average lifespan is 810 hours, with a margin of error of 35 hours.

Assuming the margins of error were calculated in the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

- A. The first sample contained fewer light bulbs than the second sample.
- B. The first sample contained more light bulbs than the second sample.
- C. The first sample had a shorter average lifespan than the second sample.
- D. The first sample had a longer average lifespan than the second sample.

Answer 13

Correct Answer: B

Question 14

A vibration model for a machine mount is given by a quartic that is written in shifted form. The function $p(x)$ is defined by:

$$p(x) = a((x - 3)^2 + b)((x - 3)^2 - c),$$

where a , b , and c are constants. The graph of $y = p(x)$ passes through the points $(1, 25)$ and $(-1, 52)$.

What is the value of $p(5) + p(7)$?

- A. 25
- B. 27
- C. 52
- D. 77

Answer 14

Correct Answer: D

Question 15

A savings balance is modeled by an exponential function of time. In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$.

Which expression gives the value of c in terms of b ?

$$f(x) = 66(b)^x$$

- A. $1 + \frac{b}{100}$
- B. $b + 100$
- C. $100(b + 1)$
- D. $100(b - 1)$

Answer 15

Correct Answer: D

Question 16

A park service tracks the average growth of a young oak. On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

A. $\frac{29m}{4k}$

B. $\frac{29k}{4m}$

C. $\frac{1044m}{k}$

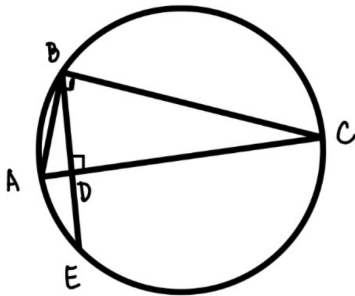
D. $\frac{1044k}{m}$

Answer 16

Correct Answer: D

Question 17

A circular plaza is being surveyed. In the figure shown, points A , B , C , and E lie on the circle, and $AB < BC$. Angle ABC is a right angle, segment AC is a diameter of the circle, and segment AC is perpendicular to segment BE at point D . Also, $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r ?



Answer 17

Final Answer: 86.5

Question 18

An engineer solves a timing equation before calibrating two sensors. In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a ?

$$(x - k)^2 = (k - 5a)(x - k)$$

- A. $-\frac{32}{5}$
- B. $\frac{32}{5}$
- C. $-\frac{5}{32}$
- D. $\frac{5}{32}$

Answer 18

Correct Answer: A

Question 19

A lab predicts bacteria population (in thousands) with an exponential model. The equation $N(m) = 63(Q)^{m/6}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds.

What is the value of p in terms of Q ?

- A. $100(Q^{30} - 1)$
- B. $100(Q^{\frac{1}{2}} - 1)$
- C. $100(Q^{30} + 1)$
- D. $100(Q^{1.5} + 1)$

Answer 19

Correct Answer: B

Question 20

A logistics planner solves an equation whose solutions represent two delivery windows. In the given equation, m and n are constants. The sum and product of the solutions of the equation are 8 and $\frac{7}{2}$, respectively.

What is the value of $\frac{m}{n}$?

$$(x - m)^2 = (m + 2n)(2x - n)$$

A. $-\frac{1}{3}$

B. $\frac{1}{3}$

C. -3

D. 3

Answer 20

Correct Answer: D

Question 21

A landscaping crew is staining a yard fence and must buy enough product for two coats. One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S , in gallons, needed to stain the fence in this yard twice?

A. $S = \frac{w}{170}$

B. $S = 170w$

C. $S = 340w$

D. $S = \frac{w}{85}$

Answer 21

Correct Answer: D

Question 22

A projectile's height in a scaled coordinate model is given by $y = -x^2 + 9x - 100$. In the xy -plane, the graph of this equation intersects the line $y = c$ at exactly one point. What is the value of c ?

- A. $-\frac{481}{4}$
- B. -100
- C. $-\frac{319}{4}$
- D. $-\frac{9}{2}$

Answer 22

Correct Answer: C

Thank You!

We appreciate your time and focus.

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